

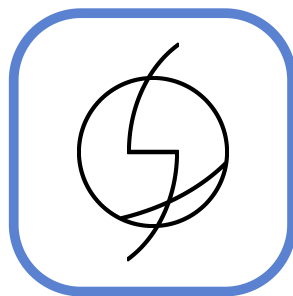
A course in language creation



Conlangs University

Phonology — Class 2

Designing a phonology



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This part of the phonology course will be concerned with what is, and how to build, "a phonology". It will deal mostly with *naturalistic* phonology, as **unnaturalistic** phenomena are dependent on this and largely ungoverned by the scope of our linguistics.

Part 3 will contain a lot more theory that you will be able to toy around with once you have defined the basics of the sounds of your language.

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Please take the time to remember that you are not documenting something that exists. You are building something, and any guidelines present in either this course or in natural languages ***can be ignored if you so wish.***

Natural languages, and our linguistics, can only provide a reference of what is attested. This course is making an attempt at exposing the broad strokes of linguistic science.

Keep in mind that "what is attested" is not "what is possible", nor is it "what you must follow".

Prerequisites

Please make sure to have read [Conlang University's Phonology 1](#) and understand how to read and use the International Phonetic Alphabet, as this document will make use of it.

It also contains an explanation of the difference between a *phoneme* and a *phone*.

A language's phonology

Most languages, including conlangs, have a phonology. This means **all spoken languages**, but we also use the term "**phonology**" for **sign languages**, as their signs are not very different from phonemes in how they're used.

A phonology, when speaking about a language's phonology, is different from the discipline of phonology in that it refers to all of **the sounds present in the language as well as the set of rules that govern the formation of syllables and words in a given language and how they are realised**.

Even further it can be said that each speaker — native or not — of a language has their own phonology for the language they speak: while the symbols chosen for the *phonemes* of two given speakers might coincide, the *allophones* of each single phoneme might not.

In other words, two speakers of the same language can have the same phonemes, but pronounce sounds differently.

Phone{m,t}ic inventories

A *phonemic* inventory is not the same as a *phonetic* inventory: as said before, *phonemes* may be realised as different *phones* depending on their context.

As such, the *phonetic* inventory of your language may be larger than its *phonemic* inventory.

Of course, when building a conlang the primary speakers are often... Just one person: the author. You. With this in mind, constructing the *phonetic* inventory of your language can seem useless, because you are going to produce your own allophones, and no other speaker would be able to prove you wrong as there would be no other speaker.

That said, there are very good (and at least one bad) reasons to want that level of details.

1. **The allophones you want to have are not those that come to you naturally**, and you want some consistency across your recordings;
2. **You aren't the only speaker**, and want to have all speakers pronounce things the same way;
3. **You are not just documenting a snapshot of the language**, taken in-between rapid sound changes, and want to have it evolve further down the line;
4. **You have a compulsive need to be as thorough and scientific as possible**, and you enjoy suffering;
5. **You want to do better than everyone else and think "more" is synonymous with "better"**.

Obviously this list is not exhaustive and you may have another reason for going into such excruciating details. Or no reason at all, you really don't need one to enjoy what you do!

Picking signs

This chapter is not presenting an exhaustive list of ways to come up with parts of a phonology, and you can always tweak any method to fit your needs and wants as much as you like. It is merely offering various starting points.

Considerations

Generally, when considering which sounds to have in the inventory of your conlang, you are faced with the following choices:

1. Choose phone(me)s you are able to pronounce and distinguish;
2. Choose phone(me)s that are common across languages;
3. Choose phone(me)s that you find aesthetically pleasing;
4. Choose phone(me)s to experiment with an idea;
5. Choose phone(me)s based on a specific input.

The reasons behind which choice you will pick can be numerous and diverse, and will largely depend on your goals.

One choice among the 5 above does not necessarily exclude the others, but it is likely that any one of them will need to be the leading one and will constrain the others.

For instance, you cannot add *all* the common phonological segments if you can not pronounce all of them.

Speaking your language

As stated in the first document of this course, many conlangers have as a goal to be able to pronounce their language.

Choice 1 will be your leading option when creating a language you want to be able to pronounce, be it for communication with friends or for fun.

On top of this, you could consider adding choice 3 to your language's design constraints, as you will probably not want to be speaking the language if you dislike the sound of it.

Naturalism & Naturalness

Choice 2 is useful if you want to maximise *naturalism*, which is the practice of staying as close as possible to what *existing, naturally evolved languages* do.

To do so, some resources are available, such as [PHOIBLE](#), a database of cross-linguistic phonological data.

Though incomplete and "only" surveying a little bit south of 2200 languages, it is still a tool that is worth pointing out and that offers a good look into *naturalness*.

Here are a few generalities (source: *Introducing Phonology 2nd Edition*, David Odden, 2013; page 207):

- Of the places of articulation represented on a standard IPA table, the places of articulation of [p, t, k] are the most common;
- Postalveolar consonants are the next most common;
- Uvular, dental and retroflex consonants are less common;
- Pharyngeal consonants are least common;
- All languages have a series of simple consonants lacking secondary vocalic articulations;
- The most common secondary articulation is rounding applied to velars, then palatalisation; relatively uncommon is rounding of labial consonants;
- Least common would be distinctive velarization or pharyngealization of consonants;
- Among consonants with multiple closures, labiovelars like [kp] are the most common;
- Clicks, though rare, seem to be more common than linguolabials;
- Nearly all languages have at least one nasal consonant,
- Most languages also have at least one of [r] or [l], and typically have the glides [w j]
- Among laryngeal glides, [h] is the most common, then [ʔ], followed by the relatively infrequent [ɦ];
- The most common vowel system would seem to be [i e a o u];
- There are fewer languages with just the vowels [i u a] than with the full set [i u e o a];
- The commonness of front rounded and back unrounded vowels is correlated with vowel height, so a number of languages have [y] and not [ø], but very few have [ø] and not [y];
- Full exploitation of the possibilities for low vowels [æ ɶ a ɒ] is quite rare, but it is not hard to find languages with [i y ɪ u];
- Oral vowels are more common than nasal vowels, and modal ("normal") voiced vowels are more common than creaky-voiced or breathy vowels.

NOTE:

These generalities are **not** rules, and twisting or breaking them is allowed!

Symmetry and gaps

Looking at a language's inventory, it is generally expected that there will not be broad categories, or classes, (place of articulation, manner of articulation, phonation) that can be said to bear only "one of" the phonemes they encompass. For instance, it isn't expected to have only one voiceless fricative in a language, or only one alveolar consonant, or only one nasal.

There are of course some exceptions to this, particularly in places of articulation that only bear a few sounds (such as the glottis, for instance).

Equally, *not* having a single distinction would also be weird: if your language has voiced and unvoiced fricatives in 4 places of articulation but only one of them in a fifth place, that's a surprising gap.

Further down this document, the section *Resources & Inspiration* features WALS articles about some more common gaps.

... in consonant systems

Symmetry in consonant systems never occurs throughout *all* of it. Instead, it is found both within and across classes of consonants:

1. **within** a same manner of articulation (for instance, every plosive exists as both voiced and unvoiced);
2. **across** two (or, sometimes, more) manners of articulation (for instance, similarities between the distributions of both plosives and fricatives).

Every language has gaps in its phonemic inventory. They may be due to the inability for humans to produce and articulate certain sounds — such as bilabial laterals (laterals imply the air flowing on the sides of the tongue, which isn't involved in the production of bilabial consonants) or glottal trills (the glottis physically cannot move this way) — or to the language simply... being that way, and doing without some sounds, or classes of sounds.

Let's look at a few inventories:

Inventory 1 (English)

↓ Moa / PoA →	Bilabial	Labio-dental	Dental	Alveolar	Post-alveolar	Palatal	Velar	Glottal
Plosive	p b			t d			k g	ʔ
Nasal	m			n			ŋ	
Affricate					tʃ dʒ			
Fricative		f v	θ ð	s z	ʃ ʒ			h
Approximant				ɹ				
Lateral approximant				l		j	w	

Inventory 2 (Malay)

Every plosive exists as both unvoiced and voiced, every PoA with a plosive has a nasal, no discrepancy in voiced/unvoiced pairs as they're only in the plosives.

↓ Moa / PoA →	Bilabial	Alveolar	Palatal	Velar	Glottal
Plosive	p b	t d	c ɟ	k g	
Nasal	m	n	ɲ	ŋ	
Fricative		s			h
Trill		r			
Lateral approximant		l			

Inventory 3 (Daakaka)

Every PoA that has plosives also has nasals. Every plosive exists as both unvoiced and voiced (albeit prenasalised). The discrepancy in voicing between the two fricatives is a tiny bit of asymmetry.

↓ Moa / PoA →	Labio-velar	Bilabial	Alveolar	Palatal	Velar
Plosive	p ^w m ^b w	p ^m b	t ⁿ d		k ^ŋ g
Nasal	m ^w	m	n		ŋ
Fricative		v	s		
Trill			r		
Approximant	w			j	

Inventory 4 (Arrernte)

Notice here that nearly every PoA/MoA intersection has a pair of phonemes in it. Some wonderful symmetry here!

↓ Moa / PoA →	Bilabial	Dental	Alveolar	Palatal	Retroflex	Dental
Stop	p p ^w	t̪ t̪ ^w	t t ^w	c c ^w	ʈ ʈ ^w	k k ^w
Nasal	m m ^w	ɳ ɳ ^w	n n ^w	ɲ ɲ ^w	ɻ ɻ ^w	ŋ ɳ ^w
Prestopped nasal	ᵐᵐ ᵐᵐ ^w	ᵗᵗ ᵗᵗ ^w	ᵗᵗ ᵗᵗ ^w	ᶜᶜ ᶜᶜ ^w	ᵗᵗ ᵗᵗ ^w	ᵗᵗ ᵗᵗ ^w
Prenasalized stop	ᵐᵇ ᵐᵇ ^w	ᵗᵇ ᵗᵇ ^w	ᵗᵇ ᵗᵇ ^w	ᶜᶜ ᶜᶜ ^w	ᵗᵇ ᵗᵇ ^w	ᵗᵇ ᵗᵇ ^w
Lateral Approximant		ɭ ɭ ^w	l l ^w	ʎ ʎ ^w	ʟ ʟ ^w	
Approximant	β			j j ^w	ɻ ɻ ^w	ɰ
Tap			ɾ ɾ ^w			

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Here, it gets a bit crazier.

Inventory 5 (Arapaho)

Okay what's going on here? There is barely any symmetry, and all we can say is that *most*, but not all, plosives and fricatives have similar places of articulation.

↓ Moa / PoA →	Labial	Dental	Alveolar	Post-alveolar	Velar	Glottal
Plosive	b		t		k	ʔ
Nasal			n			
Affricate				tʃ		
Fricative		θ	s		x	h

Notice that this language lacks both /p/ and /g/, which are the most commonly omitted plosives across languages.

Inventory 6 (Kiribati) Every nasal has a lengthened version!

... that's it for symmetry.

The quasi-absence of fricatives is also a bit unusual but fear not, for /t/ lenites to [s] before an /i/! And /β^y/ can be a flap. This has nothing to do with symmetry, but it's interesting to note.

↓ Moa / PoA →	Bilabial	Alveolar	Velar
Plosive	p	t	k
Velarised plosive	p ^y		
Nasal	m m:	n n:	ŋ ŋ:
Velarised nasal	m ^y		
Approximant/Flap	β ^y	r	

Inventory 7 (Karajá)

Don't you love it when there's only symmetry in one single place, and it's only in the most common of them, phonation?

Also note that /k/ is barely present in men's speech, though it is preserved in at least one grammatical ending.

↓ Moa / PoA →	Labial	Dental	Postalveolar	Velar	Glottal
Plosive	b	d		k	
Affricate			tʃ dʒ		
Implosive		ɗ			
Fricative		θ	ʃ		h
Lateral		l			
Sonorant	w	r			

NOTE:

To be clear, the /k/ is still present in more than just that one grammatical ending, but it's still rarer in male speech.

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One small thing to note is that there is a tendency with symmetry and asymmetry: the larger an inventory, the more chances it has to feature symmetries.

... in vowel systems

The number of phonemic vowels that any of the world's known languages can have can go from 3 to a maximum north of 20. While Quechua, classical Arabic, Greenlandic or many Australian aboriginal languages have a trivocalic system of /i a u/, Ju|'hoan can, depending on analysis, be said to sport over 30.

The three vowels [a i u] are almost always the ones, or very close to the ones, found in languages with a trivocalic system. This is because vowels are limited in space, in the oral cavity, and those three are maximally distinct from each other as:

- [i] has the tip tongue **high** and **front** in the oral cavity;
- [a] has the blade of the tongue **low** and **central** in the oral cavity;
- [u] has the back of the tongue **high** and **back** in the oral cavity.

Having only three vowels is rare, as it severely limits the number of maximally distinctive words a language can have, but adding more vowels decreases the amount of articulatory (and acoustic) distinction between the vowels, and having over 20 vowels is also rare.

So, is there an optimal number of vowels for a language to have? The answer seems to be "no", since speakers of languages with a high amount of vowels don't seem to have any problem distinguishing them. However, most languages in the world seem to have between 5 and 8 distinct vowels, with around a quarter of them all featuring the same 5-vowels system /i e a o u/ .

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Because of these space limitations described above, keeping all the vowels of a language maximally distinct from one another *requires* some symmetry.

This doesn't mean that asymmetries cannot occur in vowel systems, but it could mean that such a system would be more unstable and more prone to change.

Change is not necessarily something you would want to worry about, especially if you are not looking to **evolve** your language and are only seeking to **have** a language.

GNS

NOTE (bis):

These generalities are **not** rules, and twisting or breaking them is allowed!

SOURCES for the inventories:

[A guide to the languages of the world](#), Merritt Ruhlen, 1976

[UPSID Illustrations of the IPA: Central Arrernte](#)

Auxlangs & *a posteriori*

Both auxlangers and conlangers deriving their language from an existing one will take an interest in choice 5, where their inventory will be constrained by one or several inventor{y,ies} from (an)other language(s).

An *a posteriori* conlanger will have to take into consideration the existing phonology of the language their conlang will be derived from, and the daughter will likely not stray too far from the parent — although this can also be a design goal, in which case the changes will probably be documented and follow some logic.

An auxlanger will have to adapt the phonologies of the languages they're making an auxlang for, and the resulting inventory won't include sounds that aren't present in *any* of the languages taken as input. It is also likely that the auxlang's inventory will mostly make use of sounds common to *all* or *most* of the input languages.

For a more specific type of auxlangs, those that do not care about a specific genetic or areal relation between the languages it aspires to be an auxlang for, two choices are open.

Those with a limited set of input languages can still go for choice 5, although finding common ground will be harder as the languages won't be related.

Those that aspire to be a universal means of communication for people across all languages can lead with choice 2, picking sounds that are simply common across all languages.

Method 1 — *Just do it*

You could start by simply picking all the sounds you like, all the sounds that appear to conform to your idea of how your language would sound, all the sounds that fit your theoretical linguistic idea(s), and/or all the sounds present in your input.

Doing this, you'll likely find yourself browsing [an interactive IPA table](#) and repeating the sounds you hear, paying mindful attention to the placement and activity of your tongue. You'll look ridiculous to anyone observing, but that's alright: we've all been through this, and we all go through it again and again. It's fine.

With this method, you may end up with something utterly unnaturalistic that does not conform much to the *Naturalism and naturalness* section above. You'll likely need to tweak it a bit after you're done with the initial set-up if you want something more in line with what happens in our world's languages.

Method 2 — Imitation

This one is most useful if you want to achieve a similar sound to that of a natural language. Whichever language you set your views upon, you can probably find it on Wikipedia by searching "[language] phonology".

From there, copying the inventory would get you only one step closer to mimicking the language's sound. You will need to look into its *phonotactics*, pertaining to which sounds are allowed together and/or in what order, and its *allophony*, pertaining to how the phonemes are realised depending on their context.

Method 3 — Analysis

Either record your voice or write down, in IPA, a few sentences of your language. Use the sounds you want, and feel free to tweak it and record/write it again if you're not pleased with the result. If you did a recording, try and transcribe it as accurately as you wish.

Make a list of the sounds you used, add those you wanted but forgot to use, and put all of them in an IPA consonant table and a vowel chart.

Then, you can start breaking apart the words into syllables, and take note of how the syllables are built: what's allowed? What happened in the transcription/recording that you dislike and want to get rid of? Which order do the sounds come in? If you allow two consonants in consecutive order, are *all* combinations allowed (note that the answer is very likely "no")? Make a note of which aren't. Or which *are*, if that's faster!

From there, you *could* further the list of allowed combinations and list ALL of them, but that could get boring and quite long.

Instead, try and group the sounds that are allowed in similar positions. Are only some vowels allowed after a certain sound? Do they have anything in common (are they all rounded? Back? Open? Nasal?)? If not, they probably should: sound rules are more rarely discriminating against a *single* sound.

Phonology 3 will contain a lot of help for how to formulate these rules.

You can also apply this method to an existing lexicon if you have a language with some content but no set phonology. Or if you didn't care about sounds much and only made words up using a romanisation or the language's orthography, and now want to document the underlying rules.

GNS

You don't need to keep yourself to any single one of these methods, or even to any of them. None of them is "better" than any other, and they're not the only ones to exist, they're only those I could think of as I was writing this document.

Please do bring some nuances and subtleties to them, devise your own process!

Phonological constraints

This part will make use of some things that you will not have learned yet, and that will only be talked about in Phonology 3.

This is in order to introduce some very complex notions that I think need to be explained bit by bit, but the basics of which are needed to start creating a phonology for your conlang.

Syllables

What's a syllable?

Of all phonological entities, syllables are the ones that stand out the most to native speakers on a conscious level: people can count out the syllables in a word easier than they can count the phonemes in it.

A syllable is a purely phonological entity: it is not linked to a grammatical or semantic unit and does not have meaning in and of itself.

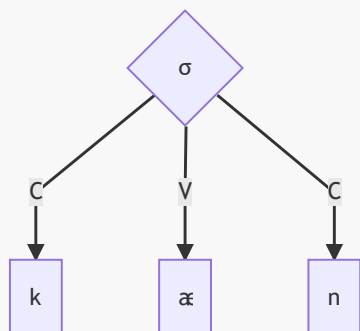
Some syllables are co-extensive (have the same boundaries) with a meaning beyond their phonological existence, but it is not inherent to it. For instance:

- the syllable [ʌn]
 - is co-extensive a morpheme in "unusual" [ʌnju:ʒəl],
 - is not co-extensive a morpheme in "gun" [gʌn];
- the syllable [kæt] is co-extensive with the word "cat" which designates a feline;
- the syllable [kæts] is co-extensive with
 - the word "cats" which designates multiple felines,
 - two morphemes *cat*, as above, and *-s*, the plural marker;
- [kæməɪl] has two syllables [kæ] and [məɪl] which are each a part of a single morpheme.

Syllable structure

Syllable structure in a given language is defined by **phonotactics**, the set of rules dealing with permissible combinations of phonemes.

Here is a simple syllable, deconstructed into its phonemes and represented in a tree. It stands for the word "can" [kæn].



You can see here that each phoneme in /kæn/ is linked to either a **C** or a **V**, which are in turn linked to the Greek letter sigma **σ**.

That sigma means "this is one syllable", and the C and V stand for Consonant and Vowel, respectively.

The CV line can be omitted, as it's generally quite clear which phonemes are Consonants and which are Vowels.



Now, let's talk about structure: what's a syllable made of?

NOTE:

This section will only present a broad overview of the principles of syllabification and syllable structure that would fit an approximation for most languages.

It would be impossible to fit all the rules that would make it possible to define syllabification in all languages into a format suited for a course in language creation.

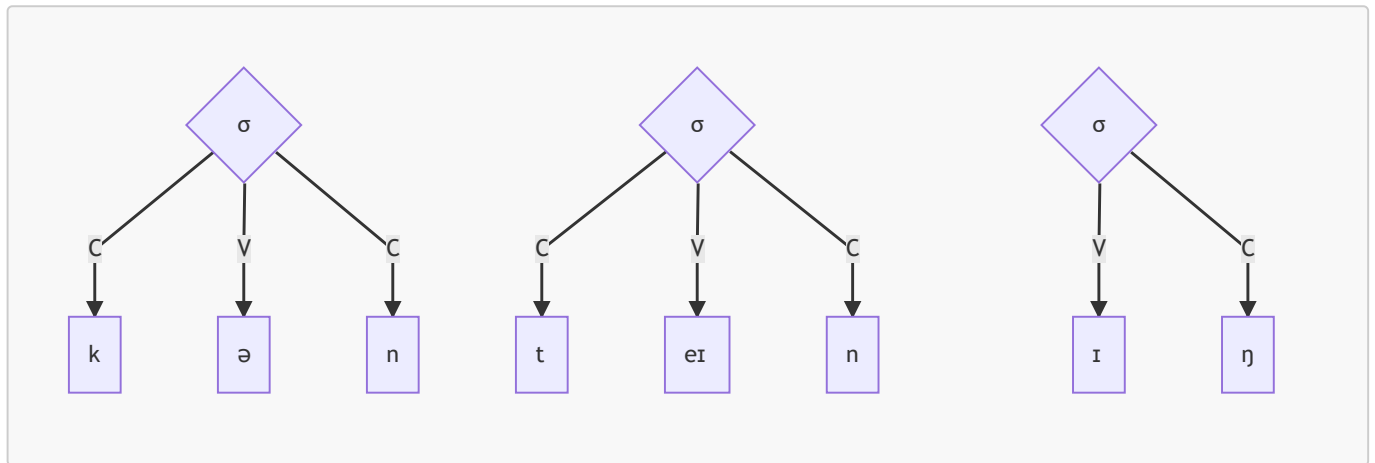
If you want to know more, you can grab a phonology textbook such as:

- *Introducing Phonology*, David Odden, Cambridge Introductions to Language and Linguistics
- *Introductory Phonology*, Bruce Hayes, Blackwell Textbooks in Linguistics
- *Phonology: An introduction to basic concepts*, Roger Lass, Cambridge Textbook in Linguistics

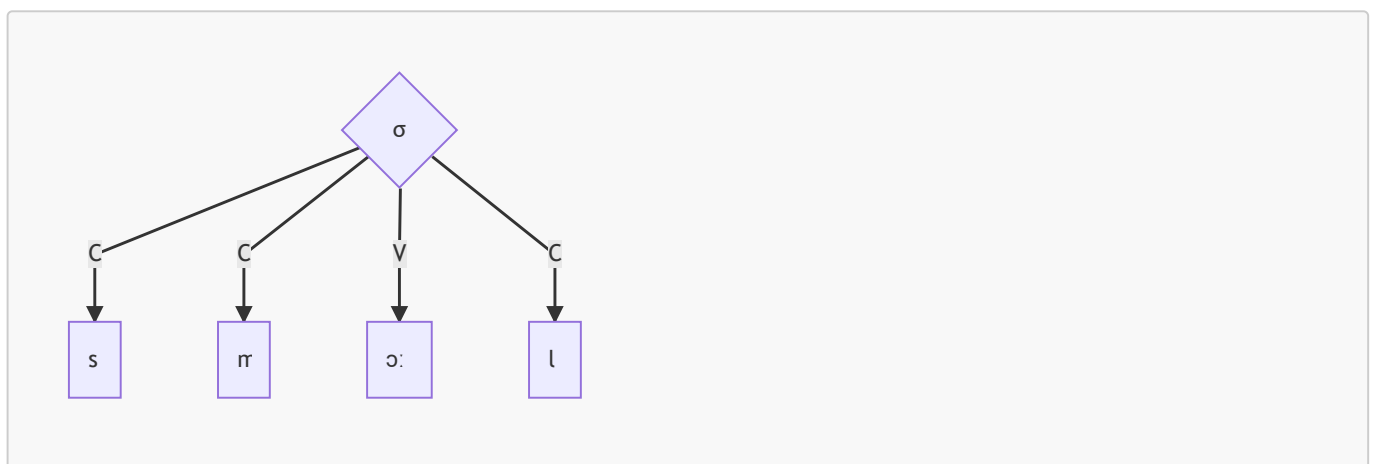
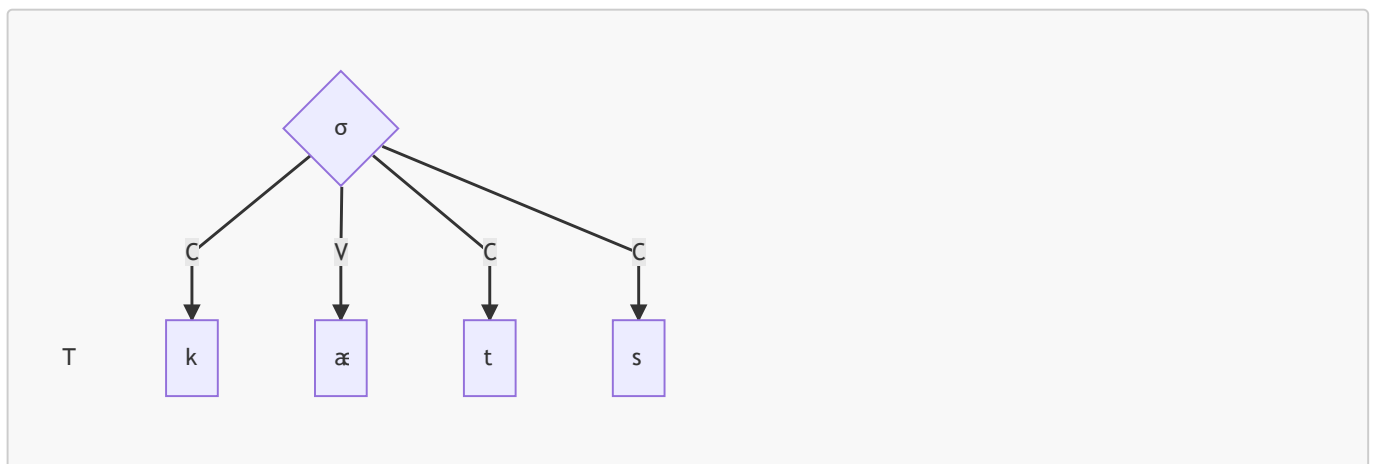
Let us consider that same syllable [kæn]:

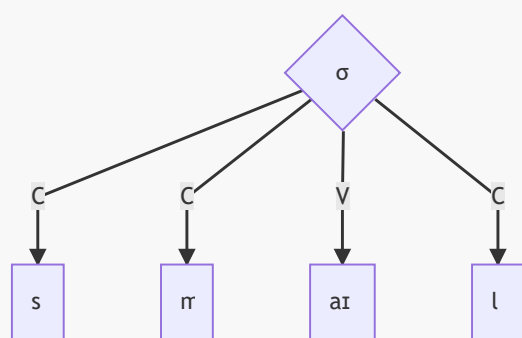
- its *onset* is **k** ;
- its *nucleus* is **æ** ;
- its *coda* is **n**

And then the same system, but with a word containing multiple syllables:



Some syllables may be described as "complex", meaning they have more than one element for either their onset, nucleus, or coda. They would be deconstructed this way, using "cats", "small" and "smile".





The σ is generally placed right above the **syllabic** component, generally the vowel (and thus noted V), but it can also be a **syllabic consonant**, hence the name. Refer to Phonology 1 for a short explanation.

Syllables can also be marked in phonemic or phonetic transcriptions by dots, as seen in Phonology 1: $['sɪ.lə.bəl]$, $[kən.'teɪ.nɪŋ]$, or using subscript sigmas before opening brackets $]_{\sigma}$ and or after closing ones $]_{\sigma}$: $[_{\sigma}kən]_{\sigma}[_{\sigma}teɪ]_{\sigma}[_{\sigma}nɪŋ]_{\sigma}$.

Isn't this absolutely, perfectly horrible? We can however simplify it a bit by only specifying where syllables end: $[kən]_{\sigma}[teɪ]_{\sigma}[nɪŋ]_{\sigma}$ is (only a little bit) less jarring, but really please don't use this in a whole word it's disgusting. I'm only telling you because some sadists use it in textbooks or grammars.

WAAAAAAAA

the tree breaks up $kən'teɪnɪŋ$ as $kən.'teɪ.nɪŋ$ but the inline notation does it as $kən.'teɪ.nɪŋ$! HAHA you made a mistake, silly you!

Okay good, you were paying some attention! This isn't a mistake: both are valid ways to break up the syllables.

The latter of those two conforms to the *Maximal Onset Principle*, or **M.O.P.** in some sources, which dictates that if an intervocalic consonant can be assigned to the onset without breaking other rules of syllable formation within the language, it is.

Sometimes, it's more relevant to preserve morphological boundaries when transcribing a word ($/kən.'teɪ.nɪŋ/$ shows that -ing is separate) and this is mostly the case in phonemic transcription ($/\ /$) as phonetic transcriptions ($[]$) can be closer to what is actually produced. [This wikipedia article about english phonology](#) has a small paragraph about this.



Nucleus

Syllables always contain a *nucleus*. It is the only cross-linguistically essential and necessary part of all syllables.

It is generally a vowel or diphthong, although sometimes a consonant can accomplish that role. Such a consonant is said to be a *syllabic consonant*. You can find some in English, in words like "bottle" $[bɒtəl]$ (the first vowel may differ depending on your dialect) or "rhythm" $[ɹɪðəm]$.

A note about diphthongs

Say you have two vowels in a row: how do you determine whether they're really two different segments or a single one (diphthong)?

The short answer is that if they're part of the same syllable, then they're a single unit, as there is a one-to-one correspondence between the number of syllabic components and the number of syllables. As such, two different vowel units would be in two different syllables.

Onset & Coda

The onset is an optional component of syllables, if you look at all languages.

But there are languages in which syllables must begin with an onset, and cannot begin in a vowel.

The coda is also optional, as not all languages require their syllables to have one. However, no language requires that all syllables have one.

While onsets are never forbidden to appear at all in any language, there are some that forbid codas.

A syllable without a coda is said to be "open". A syllable with one is said to be "closed".

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Onset and coda are where *consonant clusters* can happen. Most languages have rules about which combinations can occur.

Sonority hierarchy

Sounds can be arranged following a "sonority" scale, basically corresponding to their loudness in speech, or more technically to the amount (from highest to least) of vibration in the vocal folds. The general scale is as follows, from most sonorous to least sonorous:

- vowels
- glides
- approximants
- nasals
- fricatives
- stops

A general rule is that sounds that are *more sonorous* tend to be "internal" — close to the syllable nucleus — while *less sonorous* ones tend to be "external" — closer to the edges of a syllable.

This rule is not always strictly followed: in English for example, [s] can come before (be external to) stops even though they are *more sonorous*. Consider the monosyllabic words "stretch" or "space".

This means the pattern /**pralt**/ would be allowed as a single syllable in most languages (assuming they do contain these sounds, and allow clusters), while /**rpatl**/ would not.

Some languages would deal with the latter by making some of the consonants syllabic: [ɾ.pa.t̚l] , for instance.

This rule is also not always strictly followed. Consider for instance some monosyllables found in metropolitan French ("câble" [kabl]) or Russian ("pra" [rta], genitive singular of "por", meaning "mouth").

These features can play massive roles in the phonotactics, sometimes ruling out a lot of consonant clusters. In English for instance, initial stops may only be directly followed by approximants or glides: "play" [plei], "crowd" [kɹaʊd], "cute" [kjut] or "twin" [twin]. However, no language is known to forbid vowels after any class of consonants.

Resources & inspiration

Here are some commonly used resources for building a language's inventory and/or phonology:

- [Balancing Vowel Inventories](#) by u/xain1112
- [WALS Feature: Consonant Inventories](#)
- [WALS Chapter: Consonant Inventories](#)
- [WALS Feature: Vowel Quality Inventories](#)
- [WALS Chapter: Vowel Quality Inventories](#)
- [WALS Feature: Voicing and Gaps in Plosive Systems](#)
- [WALS Chapter: Voicing and Gaps in Plosive Systems](#)
- [John Quijada's *Phonoaesthetic considerations](#)
- [A Survey of Some Vowel Systems](#)
- [Conlangery #48 — Designing sound systems](#)

NOTE:

The *Phonoaesthetic considerations* document makes use not of IPA, but of a way to type it using characters commonly found on keyboards. It is a system called X-SAMPA [and a key to read it can be found on wikipedia](#).

Here are a few languages with odd inventories you can look up: Rotokas, Hawaiian, Northern Sami, Xhosa, Abkhaz, Danish, Mongolian, Yele, Toda, Fulani, Iau, Kaytetye, Ubykh.



A phonology generator exists, and it's called Gleb: <https://gleb.000024.org/>.

Keep in mind: its output is entirely random and not supposed to be naturalistic!